

24. A non-renewable source of energy is :

- (a) Solar energy
- (b) Petroleum
- (c) Wind energy
- (d) Biogas

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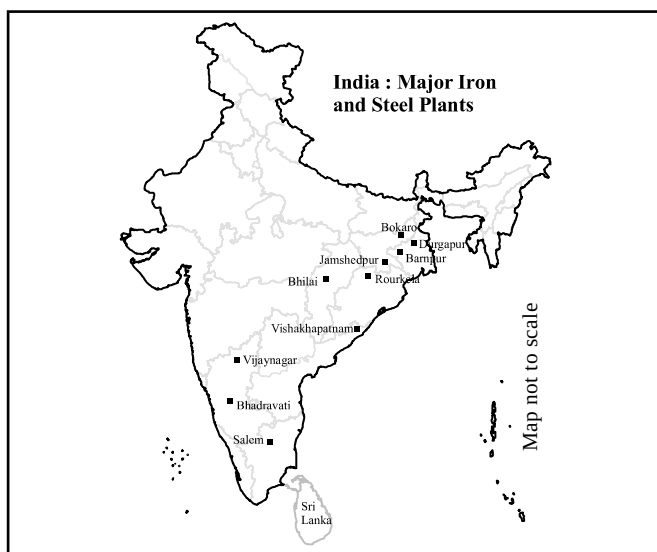
Ans. (b)

Petroleum, also called crude oil, is a **fossil fuel**. Like coal and natural gas, petroleum was formed from the remains of ancient marine organisms, such as plants, algae, and bacteria.

Industry

i. Iron and Steel Industry

*The iron and steel industry is a basic industry. The first attempt to make iron in India started in **1830 A.D.** in Porto Novo (Tamil Nadu), but this factory late failed. For the first time in India, iron was successfully produced by the Bengal Ironworks in **Kulti** in **1875 AD**. In the year 1907, a new factory was established in India by the Tata Iron and Steel Company at Sakchi (Jamshedpur). In where raw iron was produced in the year 1908 and steel was produced in the year 1912. According to World Steel Association data (World Steel in figures), 2024, it is the world's 10th largest (in the year, 2023) steel company (Tata in Steel Group, 2023) with an annual production capacity of 29.50 million tonnes.



***Visvesvaraya Iron and Steel Ltd. (VISL)** was established in the year 1923 on the bank of the Bhadravati

River in Shimoga District of Karnataka. Raw Iron is supplied from the **Bababudan Hills**, while limestone is sourced from Bhandigunda. Due to the lack of local supply of coal supply, charcoal or wood was used for power which was sourced from the **Malnad** forest area. Since 1989, it has acted as a subsidiary of the Steel Authority of India Ltd. It was completely merged with SAIL in the year 1998. *After the independence, during the second five-year plan of India, special emphasis was given to heavy and basic industries under which **Rourkela** (Odisha), **Bhilai** (Chhattisgarh), and **Durgapur** (West Bengal) steel plants were established.

***Rourkela Steel Plant or Hindustan Steel Limited (HSL)**, is located in the Sundargarh District of Odisha. This factory was established in collaboration with Germany's companies Krups and Demag. It is situated near deposits of manganese and limestone. Iron ore is brought from the Keonjhar and Kendujhar districts of Odisha. The cooking coal is obtained from Bokaro, Jharia and the Talcher area of the coal washing station in Kargali. Manganese, limestone and dolomite are obtained from Veermitrapur. It gets cheap hydroelectricity from Hirakud easily. ***The Bhilai steel plant** was established with the help of the former Soviet Union in Bhilai of Durg District in the state of Chhattisgarh on the Kolkata-Mumbai railroad. The **Dhallirajhara iron ore deposit** supplies iron ore to the Bhilai steel plant.

* **Hindustan Steel Limited (HSL)-Durgapur-** the Durgapur steel plant was established with the collaboration of Britain near Asansol in the Bardhaman district of West Bengal. This plant obtains iron ore from the Bolni mines of the Gua area, Coal from the mines of Barakar and Jharia, manganese from Jamda (Keonjhar District) and electricity and water from the Damodar River.

* **Bokaro Steel Limited (BSL) - Bokaro** -This plant was established with the collaboration of the then Soviet Union at the confluence of the Bokaro and Damodar rivers in the Hazaribagh district of the state of Jharkhand.

*Major minerals required for the iron and steel industry are iron ore, coking coal, manganese, rock phosphate, dolomite and chromium etc. Due to a lack of coking coal in the country, it is imported from Australia.

*Generally chromium and nickel are used in the making of stainless steel. Both of these are mixed with iron metal along with some quantity of carbon for the manufacturing of stainless-steel.

*The Visakhapatnam steel plant is India's first steel plant located on a coastline steel plant. Iron ore is supplied to this plant from Bailadila (Chhattisgarh).